Causal Inference

Dr Valentin Vancak, Department of Data Science
99219-1

Type of course: Class

Method of teaching: According to HIT and the department guidelines

Prerequisites: Introduction to programming with Python, Mathematics for Data Science, Statistical Methods

Credits: 3

Contact Information: valentinv@hit.ac.il

Office Hours: Wed. 15:00-16:00

Accessibility: Please contact the dean of students for any required adjustments to the course.

Course Description and Learning Objectives

Course summary

Causal inference is one of the most researched areas in data science in the last decade. The purpose of causal inference is to identify and estimate causal relationships between variables. Inference about causal relationships lies at the basis of many scientific fields and applications such as economics, epidemiology, medicine, pharmaceuticals and more. In the digital age, the modern theory of causal inference plays a decisive role in data analysis since the vast majority of collected data do not result from randomized controlled trials but rather are created naturally and in an uncontrolled fashion. Therefore, classical modelling methods may lead to biased and even contradicting conclusions regarding the effectiveness of the effect of a certain policy, intervention or treatment. In this course, we will answer the following questions: What is the difference between correlation and causation? In which cases is more data not helpful for the analysis and can even be harmful? What are the conditions and assumptions that allow for effective causal inference procedures? In which cases is the omission of explanatory variables necessary for data analysis, and in which cases is it prohibited? In addition, we will discuss parameter estimation methods and causal models.

Learning Objectives

At the end of the course, the students will be able to define, characterize and fit a causal model to the data. Additionally, the students will be able to apply different estimation methods and test the validity and sensitivity of the fitted models.

Assignment and Grading Distribution

This course requires continuous learning during the semester as the topics build on each other. During the course, 3-4 homework exercises will be given, including theoretical questions and data analysis. At the end of the course, an in-class test will be given.

Homework - 20%

In-class examine - 80%

Course Plan

No. lesson	Subject Description	Comments
1	Rubin's Causal Model (RCM) – potential outcomes, definition of a causal effect	
2	Directed Acyclic Graphs (DAGs), operations on graphs, d-separation, twin causal networks	
3	Structural equations and causal parameters	HW01
4	Classic estimation methods - regression, propensity score, IPW, g-computation, and Doubly-Robust (DR) estimators	
5	Instrumental variables - definition, assumptions and examples	HW02
6	Estimation methods based on instrumental variables in linear models (the TSLS) and in non-linear models - G-estimation (generalized estimation equations)	
7	Sensitivity analysis of estimators based on instrumental variables - strength and validity	
8	Causal mediation analysis models - definition and estimation. Nested potential outcomes. Direct, indirect and total causal effects, Sobel test	HW03
9	Pearl's mediation formula. Estimation in linear and non-linear models with mediation, proportion mediated	
10	Causal discovery from observed data	
11	Additional selected topics (effect size of causal contrasts, diff-in-diff, synthetic control) and complementary topics	
12	Course summary and review lesson	

גולומב 52 חולון 5810201, ת.ד. 305, טל': 03-5026940
52 Golomb St. Holon 5810201 Israel, Tel: +972-3-5026940
המחלקה למדעי הנתונים
Department of Data Science
www.hit.ac.il

* There may be changes throughout the semester. In any case of change, a notice will be sent in advance.

Bibliography

[1] Hernán MA, Robins JM (2019). Causal Inference. Boca Raton: Chapman & Hall/CRC, available online at https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book

[2] Pearl, Judea, and Dana Mackenzie. (2018) The Book of Why: the new science of cause and effect.

[3] Pearl, J. Causality. (2009). Cambridge University Press.

[4] Imbens, G. W., & Rubin, D. B. (2015). Causal inference in statistics, social, and biomedical sciences. Cambridge University Press.

[5] Neal, B. (2020). Introduction to causal inference. *Course Lecture Notes (draft)*.

[6] Peters, J., Janzing, D., & Schölkopf, B. (2017). Elements of causal inference: foundations and learning algorithms (p. 288). The MIT Press.

[7] Sjölander, A. Lecture slides from Introduction to causal inference course, Department of Medical Epidemiology and Biostatistics, Karolinska Institutet.

[8] Ding, P. (2024). A first course in causal inference. CRC Press

[9] Peters, J. (2017, May 10-11). *Lectures on causality* [Mini course]. MIT/LIDS event produced in conjunction with Models, Inference, and Algorithms (MIA), Broad Institute. University of Copenhagen.